

Computer Network

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Agenda

Computer Network Definition

Uses of Computer Network

Network architecture

Data transmission

Data transmission Modes and Methods

Types of Computer Network

Topologies of Computer Network

What is computer network?

What is computer network?

A **computer network** is a system of interconnected devices (such as computers, servers, printers, and other hardware) that communicate with each other to share resources, exchange data, and provide services. These devices are connected through communication mediums like wired (e.g., Ethernet cables) or wireless (e.g., Wi-Fi, Bluetooth) technologies.

Why to create computer Network?

Key Characteristics of a Computer Network:

1. **Connectivity:** Establishes links between devices to enable communication.
2. **Resource Sharing:** Allows sharing of hardware (e.g., printers) and software (e.g., files, applications).
3. **Scalability:** Can expand as new devices are added.
4. **Reliability:** Ensures data transfer is secure, accurate, and available.
5. **Efficiency:** Facilitates quick and cost-effective communication and collaboration.

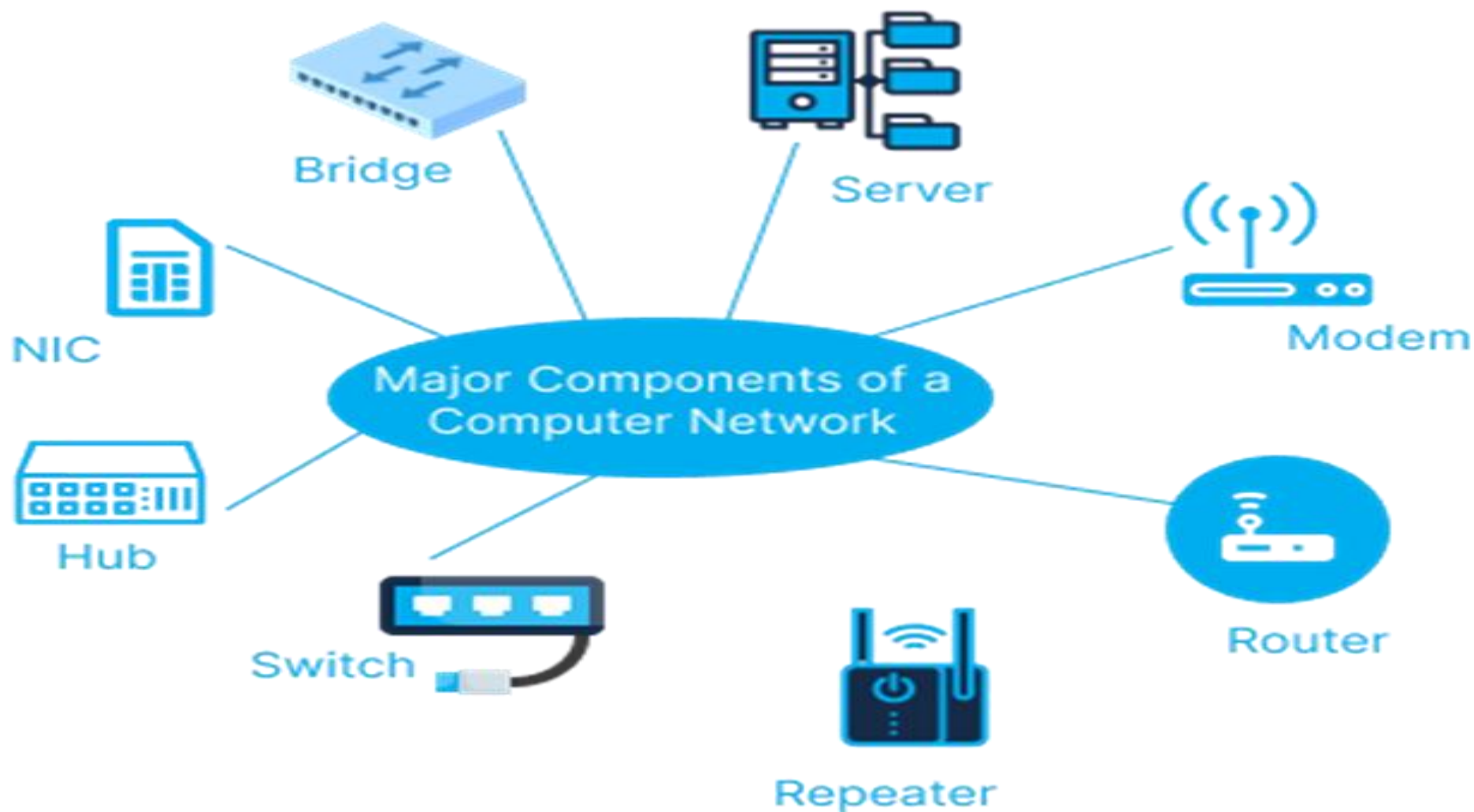
Common Uses of computer Network

Data Sharing: Sharing files, databases, and applications among users.

Resource Access: Centralized access to printers, storage devices, and servers.

Communication: Email, video conferencing, and instant messaging.

Internet Access: Connecting devices to the internet for global communication.

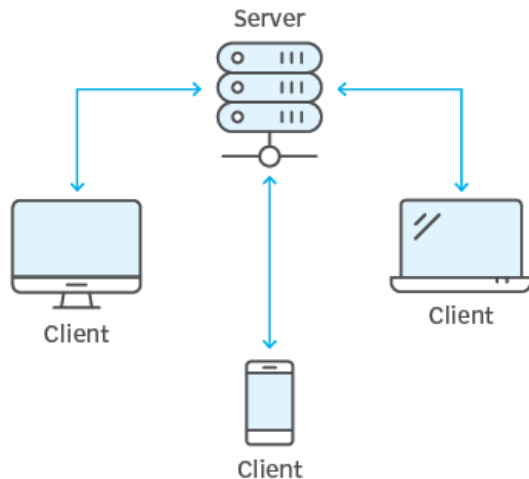


Network Architecture

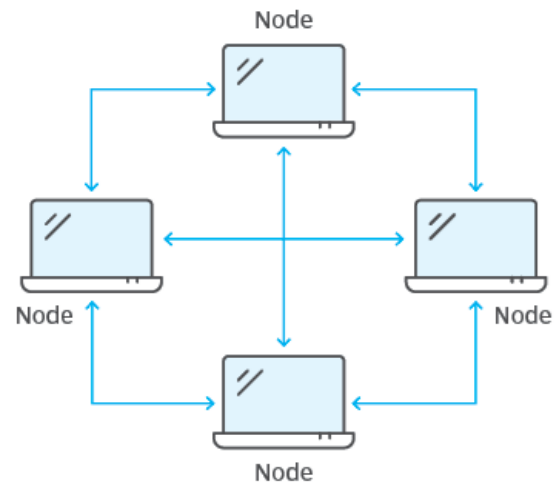
A computer network must be physically and logically designed in such a way that makes it possible for the underlying network elements to communicate with each other. This layout of a computer network is known as the *computer network architecture*.

Client-server vs. P2P

Client-server network



P2P network



Network Architecture

The following are the two most common computer network architectures:

1. **Client-server.** This model consists of many clients -- or nodes -- where at least one network node acts as the central server. The clients in this model don't share resources, but request the central server, as all the resources are installed on it.
2. **Peer-to-peer (P2P).** Each connected device on this network behaves as the client, as well as the server, and enjoys similar privileges. The resources of each peer are shared among the entire network, including memory, processing power and printing. Many companies use the [P2P](#) architecture to host memory-intensive applications, cross multiple network devices.

Data communication/ Data transmission

Data transmission is the process of transferring data between digital devices over a communication medium like wired or wireless channels. This can be categorized by the direction of flow ([simplex](#), [half-duplex](#), and [full-duplex](#)) and the method of transmission ([serial](#) or [parallel](#)).

Modes of Data Transmission

Modes of Data Transmission [?] Simplex

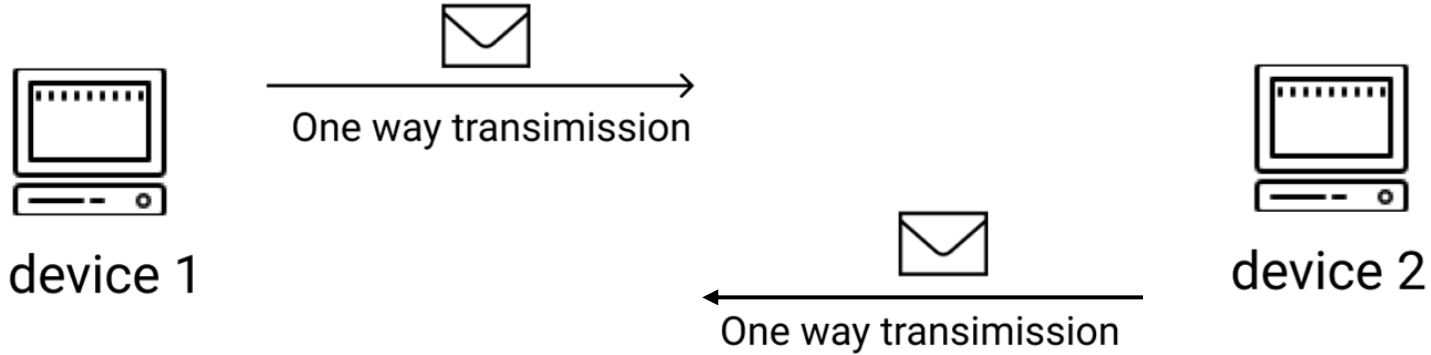


device 1

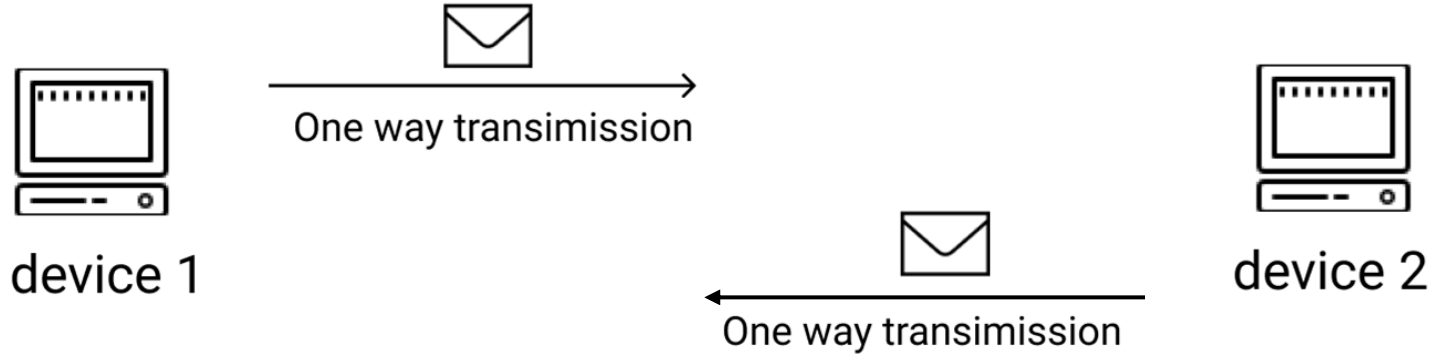


device 2

Modes of Data Transmission [?] Half Duplex



Modes of Data Transmission [?] Full Duplex



Discussion: Identify which of the following is an Example of Simplex, Duplex, Half Duplex

- Data transmission Keyboard to computer
- Data transmission of Television
- Data transmission from Mouse to computer
- Data transmission Radio Broadcasts,
- Data transmission of Monitor
- Browsing the internet (requests and responses)
- Telephone Call
- Video calls/video conferencing
- Audio calls
- Loud speaker
- A two-way radio that has a push-to-talk button
- Universal Serial Bus (USB)

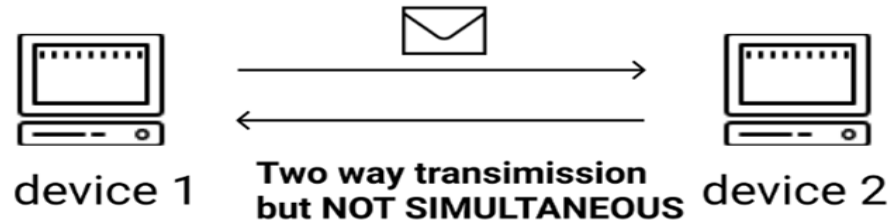
Modes of Data Transmission □ Simplex

- Simplex mode occurs when data can be sent in ONE DIRECTION ONLY
- (for example, from sender to receiver).
- Example: sending data from a computer to a printer



Modes of Data Transmission □ Half Duplex

- Half-duplex mode occurs when data is sent in BOTH DIRECTIONS but NOT AT THE SAME TIME
- *(for example, data can be sent from 'A' to 'B' and from 'B' to 'A' along the same transmission line, but they can't both be done at the same time).*
- Example: **walkie-talkie** where a message can be sent in one direction only at a time; but messages can be both received and sent.



Modes of Data Transmission □ Full Duplex

- Full-duplex mode occurs when data can be sent in BOTH DIRECTIONS AT THE SAME TIME
- *(for example, data can be sent from 'A' to 'B' and from 'B' to 'A' along the same transmission line simultaneously).*
- An example of this would be a **broadband internet connection**.



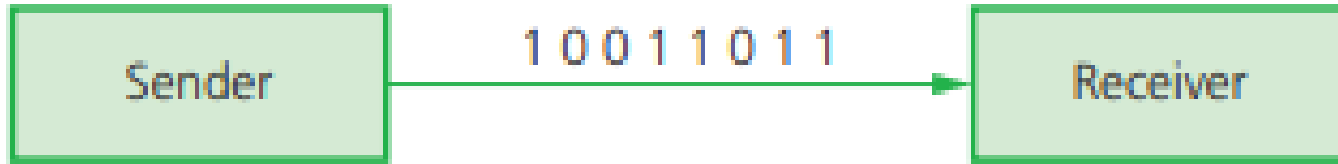
Methods of data transmission

Serial Data transmission

Parallel Data transmission

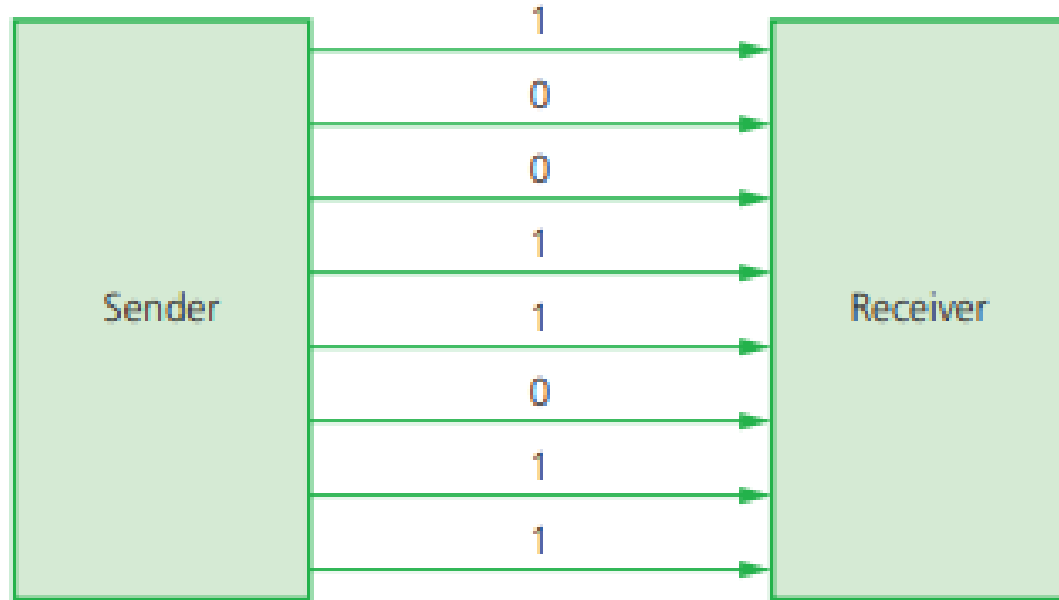
Methods of Data Transmission ? Serial

What comes in your mind when you see this picture?



Methods of Data Transmission ? Parallel

What comes in your mind when you see this picture?



Methods of Data Transmission ☐ Serial

- Serial data transmission occurs when data is sent ONE BIT AT A TIME over a SINGLE WIRE/CHANNEL. Bits are sent one after the other as a single stream.
- (Note: Serial data transmission can be simplex, half-duplex or full-duplex.)

Methods of Data Transmission ☐ Parallel

- Occurs when SEVERAL BITS OF DATA (usually one byte) are sent down SEVERAL CHANNELS/WIRES all at the same time. Each channel/wire transmits one bit:
- (Note: Parallel data transmission can be simplex, half-duplex or full-duplex.)

Activity: Define the following terms

Serial, Simplex Data Transmission

Serial, half duplex Data Transmission

Serial, full duplex Data Transmission

Parallel, Simplex Data Transmission

Parallel, half duplex Data Transmission

Parallel, full duplex Data Transmission

Recap of Computer Network

A **computer network** connects multiple devices to share data and resources; it is used for communication, file sharing, and internet access, and its **architecture** defines how devices are structured and communicate (e.g., client-server or peer-to-peer).

Recap of Data Transmission

Data transmission refers to the process of transferring data from one point to another, typically over a communication medium. This can occur between devices, such as computers, servers, or smartphones, and can happen through various methods, including:

It can be either over:

- a short distance (for example, computer to printer)

- or over longer distances (for example, from one computer to another in a global network).

Essentially, three factors need to be considered when transmitting data:

- » the direction of data transmission (for example, can data transmit in one direction only, or in both directions)
- » the method of transmission (for example, how many bits can be sent at the same time)
- » how will data be synchronised (that is, how to make sure the received data is in the correct order)

Computer Network Part II, Next Lecture

Types of Network

- a. **nano network**
- b. **Body Area Network (BAN)**
- c. **Personal Area Network (PAN)**
- d. **Near-me-Area Network (NAN)**
- e. **Local Area Network (LAN)**
- f. **Wide Area Network (WAN)**
- g. **Campus Area Network (CAN)**
- h. **Metropolitan Area Network (MAN)**
- i. **Internet Area Network (IAN)**
- j. **Interplanetary Internet (IPN);**

Network topology



Star



Ring



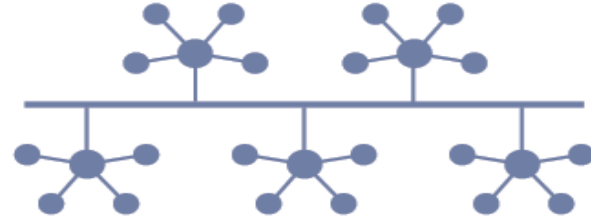
Token ring



Mesh



Bus



Tree